

Probing Universality

Replacing the Turing Yardstick with a Relative Property for Interactive GSLTs

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Abstract

Expressiveness comparisons between computational models are conventionally anchored to a single absolute yardstick: Turing completeness. For functions this works. For interactive systems it does not, because Turing completeness is a statement about the input/output relation and the input/output relation is the coarsest invariant in the linear-time/branching-time spectrum, while the equivalences that matter for interaction sit at the other end. Calculi of identical Turing power are routinely separated by uniform-encoding criteria, so the yardstick is measuring something other than what is being compared.

The obvious repair — demand behavioural rather than computational universality — merely relocates the problem, since the arbiter of which behavioural equivalence to use is an adequate modal logic, and the logical apparatus available in the ambient setting is inherited from the λ -calculus. An absolute notion is quietly λ -relative.

This note replaces the yardstick with a relative property. One fixes a probe — Turing machines, λ , CCS, π , rho, or anything else — and asks what that probe can see of the theory under investigation. No logic is required: bisimulation is identification in a final coalgebra, and the only primitive observation is reduction. What must be supplied instead are conditions preventing the question from having a trivial answer.

We recall the naive category of graph-structured lambda theories with bisimulation-preserving morphisms and exhibit three defects: the constant map to 0 is a morphism; the canonical encoding of the λ -calculus into π is not a map of terms at all, so signature preservation does not typecheck; and ambient contexts in a richer theory observe strictly more than the encoded contexts of a poorer one. We then rebuild the definition with contexts as the primary structure. A theory presents a symmetric multicategory of contexts whose interfaces record sort, binding stage, and interaction surface; transitions are labelled by contexts; and a morphism of theories is a pseudofunctor of context multicategories, so that terms, being the nullary contexts, are encoded by the same map that encodes contexts. Bisimulation on the target is computed only over the image of the probe's contexts.

Two conditions are then isolated: *hosting* (faithfulness), which says the target can run the source, and *exhausting* (density), which says the target is nothing the source cannot assemble. They induce a preorder whose degrees refine Turing completeness, the classical notion being recovered as the trace collapse — which above the Turing threshold has exactly one degree where the branching-time order has many. Our main structural observation is that the two conditions are the two halves of mutual meta-circularity: hosting yields an interpreter for the source written in the target, exhausting an interpreter for the target written in the source, and their two composites are meta-circular interpreters, each behaviourally the identity and syntactically not.

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1 Introduction

Ask whether one computational model is at least as expressive as another and the answer is normally routed through a single absolute standard. Both models are compared to the Turing machine; if both clear the bar, they are equi-expressive. That procedure is sound for models of function computation, and it is the reason the classification of programming languages by expressive power is so short a list.

For interactive systems it fails, and it fails for a structural reason rather than a technical one. Turing completeness is a statement about which partial recursive functions a model realises — a property of its input/output relation. The input/output relation is the coarsest invariant in the linear-time/branching-time spectrum. The equivalences that matter for interacting systems — bisimulation and its refinements — sit at the opposite end. A hypothesis

stated at the trace level cannot control a conclusion stated at the branching level. Concretely: an interpreter must decode before it acts, decoding is silent work, and silent work relocates choice points, so an encoding may be functionally perfect and behaviourally wrong. The gap is not hypothetical. The separation literature exhibits pairs of calculi of identical Turing power between which no encoding satisfying reasonable uniformity criteria exists [9, 10, 11].

The natural repair is to demand of a model not computational universality but *behavioural* universality: a self-interpreter faithful up to bisimulation. But stated absolutely, this only relocates the difficulty. Which bisimulation? The usual arbiter is adequacy of a modal logic, and the logical constructions available in the ambient setting are inherited from the λ -calculus. An absolute behavioural notion is quietly λ -relative, which is precisely the parochialism one was trying to escape.

This note therefore replaces the yardstick rather than recalibrating it. One fixes a *probe* — Turing machines, λ , CCS, π , rho, or anything else — and asks what that probe can see of the theory under investigation. Universality becomes a relation, not a predicate. No logic is required: bisimulation is identification in a final coalgebra, and the only primitive observation is reduction. What must be supplied instead are conditions preventing the question from having a trivial answer, and most of this note is about finding the right ones. They turn out to be two, they are standard categorical properties rather than a checklist, and together they say precisely that source and target can write meta-circular interpreters for one another.

The classical notion is not discarded. It reappears in Section 5 as the trace collapse of the relative order — a single degree sitting above a threshold, where the branching-time order has many. Turing completeness is thus the shadow of the relative property, and the separations that the classical notion cannot see are exactly the ones the shadow loses.

Outline. Section 2 fixes the class of theories through a three-rung ladder, spells out the shape of the core rewrite rule, shows why the machine models require impedance matching to enter the category at all, and explains via the McBride derivative why equations on the interaction constructor force surfaces to be reified — which matters because the contexts whose derivative is at issue are exactly the labels of the transition systems of Section 3. It closes with the defects of the naive notion of morphism. Section 3 rebuilds the category with contexts as the primary structure. Section 4 isolates the two conditions. Section 5 relates the resulting order to Turing completeness. Section 6 proves the two-interpreter theorem. Section 7 lists what is not yet settled.

2 The naive setting and its defects

2.1 A ladder of theories

The theories in play are pinned down in three stages, each rung adding structure and each still accommodating the usual suspects. The progression matters for what follows: the constructions of Sections 3–4 need only the second rung, while the interpreters of Section 6 need the third, and knowing which is which is what keeps the hypotheses honest.

Rung one: graph-structured lambda theories.

Definition 2.1 (GSLT). A *graph-structured lambda theory* G consists of a sorted signature with binding; a set of terms generated by that signature; a structural congruence $\equiv$ making the term set a graph-like quotient, so that terms are considered up to the commutative-monoid

laws of parallel composition, scope extrusion, and the like; and a rewrite relation $\rightarrow$ on the quotient.

At this rung almost everything qualifies — λ , CCS, π , rho, the ambient calculus, and Turing machines with $\equiv$ trivial. That breadth is the problem: Definition 2.1 constrains the *shape* of a theory's dynamics not at all, so nothing can be said uniformly about how one theory simulates another.

Rung two: interactive. The second rung asks only for a *site of interaction*. Following [1, Def. 2.1]:

Definition 2.2 (Interactive GSLT). G is *interactive* if it carries, over and above the data of Definition 2.1, a distinguished interacting sort P of programs; a binary *interaction constructor* $K(P, E)$ representing the interaction of a program with an environment E , an operand of the interacting sort, with $E = P$ in the process-calculus instances; and at least one base rewrite rule whose left-hand side features K , generating a labelled transition system on P modulo $\equiv$ on which bisimulation is the intended equivalence.

For the λ -calculus K is application; for CCS, π , and rho it is parallel composition; for the interaction categories it is composition along a shared interface. Nothing about surfaces or continuations is asserted at this rung. What a bare GSLT lacks and an interactive one supplies is exactly a named site of interaction and a rule that fires there.

Remark 2.3 (Where the machine models fail). The requirement that E be an operand of the interacting sort is not a technicality. A Turing machine does have an intuitive cut — between the automaton and the tape — but automaton and tape are of different kinds, so there is no binary K on the interacting sort of which the transition rule is an instance. Turing machines therefore admit no interaction-cut presentation naively, and neither do Mealy or Moore machines, whose cut is between a transducer and a stream. They are GSLTs; they are not interactive ones. Section 5 takes this seriously: the classical yardstick is not merely a coarse probe, it is not natively an object of the category at all, and must be *presented* as one — by a choice, such as encoding the tape as a process, that is in no way canonical.

Rung three: continued. The third rung decomposes both sides of the cut. It is here, and not before, that surfaces and continuations appear.

Definition 2.4 (The interaction cut). An interactive GSLT is presented in *interaction-cut form* when its generating rule is written

$$K(K_p(I, C_p), K_e(J, C_e)) \rightarrow K'(C'_p, C'_e), \quad (C'_p, C'_e) = \text{compute}(I, J, C_p, C_e) \quad (\dagger)$$

naming the interaction constructor K ; two introductions $K_p(I, C_p)$ and $K_e(J, C_e)$, each separating an *interaction surface* (I, J — the part that must match for the cut to fire) from a *continuation* (C_p the program continuation, C_e the environment continuation); the partial *contraction* compute , which on a surface match combines the continuations into residuals; and the constructor K' repackaging them.

compute is Milner's pseudo-application: applied to $\lambda y.P$ and $[u, Q]$ it returns $P\{u/y\} \mid Q$, factoring into the substitution by which the datum migrates from the environment continuation into the program continuation through the binder, and the release of the environment's residual. Binding is one mode of migration, not a feature of the cut as such; CCS realises the shape with pseudo-application degenerating to pure release.

Definition 2.5 (Surfaces and nearness). Surfaces inhabit a sort *Surf*. When K is non-free the match condition is carried by a *nearness operator*

$$\text{near}(I, J),$$

partial, and returning — when defined — the surface at which I and J meet. Interaction is gated on its being defined.

Nearness is thus *surface-valued* rather than boolean: two introductions do not merely match or fail to match, they produce a surface, and that surface locates the resources the cut may draw on. It specialises across the spectrum as complementarity of actions in CCS, name-equality of subjects in π and rho, and interface agreement in the interaction categories.

Definition 2.6 (Continued interactive GSLT). An interactive GSLT is *continued* when equipped with

- (i) a presentation of its dynamics in interaction-cut form $(\dagger)$, naming K, K_p, K_e, K' and the partial contraction compute; and
- (ii) a *section*: a computable canonical-form function on the interacting sort, i.e. a computable section of the $\equiv$ -quotient.

ciGSLT is the category of these, with morphisms the theory maps that are bisimulation-preserving on the interacting sort.¹

The adjective names a strict enrichment, not a restriction: (i) and (ii) are data added on top of an interactive GSLT, and forgetting them recovers it. It records that the theory carries the structure a construction *over* it will need, not that any such construction has been performed.

Remark 2.7 (Two carriers of the surface, and why λ survives). Definition 2.6 appears to exclude the λ -calculus, which has no sort of surfaces and no nearness operator. It does not, because a surface may be carried in either of two ways.

- (a) *Structurally, when K is free*. Position is then unforgeable: nothing drifts into contact with what it does not already neighbour. The surface is not recorded by any explicit I or J but by the structural context $K([], S)$, and *near* is trivially defined — being in the hole is being *near*. λ is exactly this case on the environment side: App is free, and its rigidity supplies the surface that no explicit J records. A nominally absent surface is not a missing match condition but one carried by the geometry of K .
- (b) *Nominally, when K is non-free*. Any equation on K loosens position: associativity rebrackets adjacency, a unit law inserts or deletes neighbours, and associative-commutativity is the limit at which no positional structure survives at all — the freely mixed bag, where rho with $K = |$ lives. Adjacency having become cheap and non-exclusive, taking proximity as authority is ambient authority and is a leak. The surface must therefore be reintroduced nominally, supplied by *near*.

The two carriers are one notion seen twice; only the carrier changes, from the geometry of K to an explicit operator. A free K and a freely mixed K are the poles, and every K carrying equations sits on the nominal side between them.

The instances differ in how much information migrates between environment and program under contraction, and this migration spectrum is what makes the class wide rather

¹The definition of a continued interactive GSLT in [1, Def. 6.1] carries a third clause, required there for a construction not treated in this note. Nothing below depends on it, and we omit it.

Instance	Surfaces I, J	Prog. cont. C_p	Env. cont. C_e	Contraction compute
CCS	$a, \bar{a}$ (compl.)	P	Q	pure release $P \mid Q$
ρ / π (sync.)	x, x (name-eq.)	$\lambda y. P$	$[Q_1, Q_2]$	$P\{Q_1/y\} \mid Q_2$
ρ / π (async.)	x, x (name-eq.)	$\lambda y. P$	$[Q, -]$	$P\{Q/y\}$
λ	$x, -$ (K_e degen.)	$\lambda x. M$	N	$M\{N/x\}$
Ambients	n (open n vs. $n[\cdot]$)	P	Q	boundary dissolve / move
Interaction cats.	shared interface	process	process	interface composition

Table 1: Anchoring instances across the migration spectrum, after [1, Tab. 1]. A surface entry of $-$ denotes a nominally degenerate surface, carried structurally by the rigidity of K rather than absent; an environment-continuation entry of $-$ denotes a degenerate continuation, which is what distinguishes asynchronous from synchronous output. Four migration modes appear: none (CCS, interaction categories), binding (ρ , π , λ), and spatial restructuring (ambients), where the contraction relocates or dissolves a boundary rather than substituting.

than narrow. CCS realises the shape with no migration at all and is in that sense the paradigmatic instance; π and ρ realise it with migration by binding; β -reduction with migration and no environment residual — the environment introduction has a degenerate surface and its continuation carries no residual, so the argument is its own continuation and pseudo-application is ordinary application.

Remark 2.8 (Which clause this note uses). Sections 3 and 4 need rung two, together with clause (i) only insofar as surfaces sharpen the interfaces of Definition 3.3; a reader who drops the surface component throughout loses resolution in the labels and nothing else. Clause (ii), the section, is what Section 6 needs: it is what makes the interpreters exist as terms rather than as relations.

Proposition 2.9 (The section makes $\text{Ctx}(G)$ effectively presented). *If G is continued then $\text{Ctx}(G)$ admits codes: equality of contexts is decidable on representatives, and hole-filling is computable.*

Proof. Clause (ii) of Definition 2.6 picks a canonical representative of each $\equiv$ -class, so a context is presented by a representative term with labelled holes. Hole-filling is a syntactic substitution on representatives followed by re-application of the section. Stage and surface data (Definition 3.3) are finite and carried along. $\square$

Proposition 2.9 is why clause (ii) cannot be dispensed with. Without it, $\text{Ctx}(G)$ still exists as a mathematical object — quotient multicategories are unproblematic — but nothing about it is nameable, and the interpreters of Section 6 would exist only as relations between terms rather than as terms of the calculi themselves.

The usual suspects. Each rung retains the examples that matter, and loses exactly the machine models.

Theory	GSLT	Interactive	Continued	K , and the section
Turing machines	✓	—	—	cut is automaton/tape: different kinds
Mealy, Moore	✓	—	—	cut is transducer/stream: different kinds
λ (β)	✓	✓	✓	App; de Bruijn form, no channel constructor needed [1, §5.2]
CCS	✓	✓	✓	; monoid normal form
π	✓	✓	✓	; as CCS plus α -normalisation
rho	✓	✓	✓	; # fuses section and quotient
Interaction cats.	✓	✓	✓	composition along an interface

The two exclusions are the point of the rung, not an embarrassment of it. What the machine models lack is not computational power but a cut between operands of the interacting sort, and it is precisely that absence which makes them unable to say anything about how one interacting system simulates another. That λ passes clause (ii) without possessing anything like rho's $@\cdot$ is the sharp test of the same point from the other side: what is required is a section, not a channel constructor [1, §5.2].

Remark 2.10 (The ladder is not the probe order). Ascending the ladder narrows the class of theories under discussion; it says nothing about their relative expressiveness. λ and rho are both continued, and Section 5 will place them at very different degrees. The ladder fixes the category; the probe order fixes the ordering within it.

Throughout, $\Downarrow$ denotes a fixed primitive observation on terms. We take $P \Downarrow$ to mean that P has a reduction, or — when a success predicate is available — that P reports success. This is the only observation we assume, and it is intrinsic to the rewrite relation. In particular no modal logic is used anywhere in this note.

Definition 2.11 (Naive morphism). A *naive morphism* $f : G \rightarrow H$ is a function on terms satisfying $P \approx_G Q \implies f(P) \approx_H f(Q)$, where $\approx$ denotes the intended weak bisimulation of each theory. Write $\mathbf{GSLT}_{\text{naive}}$ for the resulting category.

This is the definition implicitly in play when one says that morphisms of $\mathbf{ciGSLT}$ are “bisimulation-preserving and quote-faithful”. Quote-faithfulness is an extra condition specific to reflective theories and does not repair what follows.

2.2 Impedance matching: why the machine models fail

Remark 2.3 asserted that Turing, Mealy, and Moore machines are GSLTs but not interactive ones. The claim is worth working, both because the classical yardstick is built from them and because the repair — which is always available — turns out to introduce exactly the pathology this note exists to diagnose.

They are GSLTs. A Turing machine presents as a GSLT without difficulty. Take sorts for control states, tape alphabet, and configurations; terms of the configuration sort are triples $\langle q, w, i \rangle$ of state, tape contents, and head position; $\equiv$ is trivial; and $\rightarrow$ is the transition relation. Everything Definition 2.1 asks for is present. Mealy and Moore machines present similarly, with configurations pairing a state with an input stream.

They are not interactive. Definition 2.2 asks for a binary $K(P, E)$ whose operands both inhabit the interacting sort, and for a rule headed by K . The natural cut in a Turing machine is between the automaton and the tape — but the automaton inhabits the sort of control states and the tape inhabits the sort of words, so any K pairing them is a constructor of a heterogeneous product, not a binary operation on one sort. The asymmetry is of *kind*, not merely of role: in an interaction cut program and environment are interchangeable in kind while distinguished in role, whereas a tape can never be an automaton. Consequently the transition relation is not a K -headed rule closed under K -contexts, and the labelled transition system it generates does not live on an interacting sort.

Mealy and Moore machines fail for the same reason and for a second one besides. Their dynamics is a function on state and input rather than a rewrite of a term; there is no term algebra on which the step is congruent, so even granting a cut there is nothing for it to be closed under. The failure is a failure of compositionality before it is a failure of typing.

Impedance matching, and what it costs. The repair is standard and always available: reflect the environment into the interacting sort. Encode the tape as a process — a chain of cells communicating with a head process — and both operands of $K = |$ become processes. Call the result a *process presentation*, $\text{TM}^\sharp$.

Three observations, and the third is the one that matters here.

1. The encoding is a choice. Cells may be a linked list, an array of indexed agents, or a pair of stacks; each yields a different $\text{Ctx}(\text{TM}^\sharp)$ and hence a different probe. Nothing internal to the machine model selects among them.
2. The encoding is not conservative on steps. One machine transition becomes several cut firings — request, reply, commit — so the induced transition system is not the machine's but a refinement of it.
3. Those extra firings are administrative, and administrative steps relocate choice points. That is precisely the mechanism by which a functionally correct interpreter becomes behaviourally wrong (Section 1).

So the admission ticket that lets the classical yardstick into the category introduces the very pathology the yardstick is constitutionally unable to see. This is not an argument against the machine models; it is an argument against using them as the standard against which interacting systems are measured.

2.3 Locations as labels: the derivative and the reification of surfaces

Why this subsection is here. Section 3.3 will label transitions by *contexts*: a term's behaviour is recorded not by abstract actions drawn from some separately postulated alphabet, but by the contexts that can make it move. A context is a location. So the labels of every transition system in this note are locations, and a labelling is informative only to the extent that its labels are distinguishable. The question of when a location is a genuine coordinate on a term is therefore the question of how much resolution the probe order of Sections 4–5 has at its disposal, and it is settled here, before any of that machinery is built. The chain is short and worth stating in advance:

contexts separate positions	$\iff$	plug is injective
	$\implies$	context labels carry information
	$\implies$	the induced bisimulation is fine.

Everything below is an analysis of when the first line holds, what happens when it fails, and what must be added to the theory to compensate.

Remark 2.7 said that a free K carries the surface structurally, as the context $K([], S)$, and that equations on K force the surface to be carried nominally instead. The erosion can now be stated exactly, and the right instrument is McBride’s derivative [12].

Location is a zipper, and a zipper is not a term. Let the terms be $T = \mu X. F(X)$ for a signature functor F . The one-hole contexts are not ∂F but paths of layers: writing $\partial F(T)$ for a single layer with one slot vacated and the rest filled with terms,

$$\text{Ctx}_T = \text{List}(\partial F(T)), \quad \text{plug} : \text{Ctx}_T \times T \rightarrow T.$$

This is Huet’s zipper [13] read algebraically via McBride’s derivative [12]. Two points must be kept apart, and running them together is the error this subsection is written to avoid. First, Ctx_T and T are *different types*. Second, a location is a coordinate on a term exactly when plug is injective — distinct contexts must name distinct holes — and that is a statement about plug, not about any isomorphism between Ctx_T and T .

Layer derivatives for the constructors at issue. The right-hand column says what the derivative *is*; it does not say that any of these is a surface.

prefix	$F_0(X) = A \times X$	$\partial F_0(X) = A$	an action
restriction	$F_0(X) = \mathcal{N} \times X$	$\partial F_0(X) = \mathcal{N}$	a name
abstraction	$F_0(X) = \text{Var} \times X$	$\partial F_0(X) = \text{Var}$	a variable
free binary K	$F_0(X) = X \times X$	$\partial F_0(X) = X + X$	a sibling, tagged
AC K	$F_0(X) = \text{Bag } X$	$\partial F_0(X) = \text{Bag } X$	the rest of the bag

Only the two K -rows bear on surfaces at all, since only K is the site of the cut. The abstraction row is listed because it is what makes erasure by binding possible below, and because the variable it produces is easily and wrongly taken for a surface.

Free K : the derivative is a coordinate system. For a free binary K the layer derivative is $T + T$ — a sibling subterm together with a tag saying which side the hole is on — and it is the *tag*, not the sibling, that carries the positional information. plug is injective, so Ctx_T is a coordinate system on terms, and $K([], S)$ is a context that names its position uniquely. This is the precise content of position being unforgeable: the surface is the location, and no reification is needed because Ctx_T is already doing the work.

Remark 2.12 (The surface of a β -redex, and what the variable is instead). The λ -calculus is the case to state carefully, since it is the one where the surface is easiest to misidentify. $K = \text{App}$ is free, so by the paragraph above the surface is supplied by the location. An application comprises a location together with a function and an argument; the *location is the surface*, the function is the program and the argument is the environment — or, dually, the argument is the program and the function is the environment. The location itself is symmetric; which operand is assigned which role is an orientation, not something the syntax fixes. What is not available in either orientation is a nominal surface, precisely because App is rigid.

A bound variable is therefore *not* a surface. It is nonetheless closely related, and the relation is worth naming, because both are derivative data. The surface is the location at which program and environment come into contact — the ∂ of the K -layer. The variable is the name of the locations *inside* the program continuation at which the environment’s contribution will be delivered: the occurrences of x in M are a multiset of holes in M , and $\lambda x.M$ is exactly that multi-hole context internalised as a term with x naming its holes. Contact

site and delivery site are both locations, at different levels of the term and playing different roles in the cut.

This also sharpens Table 1, whose λ row records the program-side surface as x following [1, Tab. 1]. Read through the rigidity analysis, that entry is naming the delivery site rather than the contact surface; the contact surface on *both* sides is the App location, which is why the environment-side entry is nominally degenerate rather than absent.

Equations erode injectivity. Once K carries equations the derivative is computed in the quotient, and what degrades is injectivity of plug — not the type of contexts.

- *Associativity.* Trees collapse to lists and formerly distinct bracketings of the same neighbourhood become the same context.
- *Units.* Neighbours may be inserted or deleted, so one location is reachable from strictly more decompositions.
- *Associativity with commutativity.* Lists collapse to bags and $\partial(\text{Bag } T) \cong \text{Bag } T$: the derivative has forgotten *which* occurrence was vacated. For a bag containing n copies of a subterm there are n decompositions no context distinguishes, and plug is radically non-injective.

At the AC pole the zipper has ceased to be a coordinate system. This is what “adjacency has become cheap and non-exclusive” amounts to formally: not that positions are hard to tell apart, but that Ctx_T no longer separates them.

Read through the chain above, the consequence is not merely that authority leaks. It is that the labels of the transition system collapse. Two contexts that ought to witness different behaviour become the same label, so the induced bisimulation identifies terms it should distinguish, and every probe built from the theory inherits the loss. Equations on K are thus not a local nuisance about capability; they are a global reduction in the resolution of everything the theory can observe or be observed by.

Two erasures, only one of them benign. It is tempting, having computed $\partial(\text{Bag } T) \cong \text{Bag } T$, to say that context and term have become the same thing. They have not, and the temptation conceals a genuine distinction between two ways the type barrier can come down.

Erasure by binding. In a calculus with a binder and a substitution, a one-hole context is representable *as a term*: $C[\] \mapsto \lambda x.C[x]$, with plug realised by substitution. Here Ctx_T genuinely embeds into T (or into an abstraction sort over T), the embedding is faithful, and nothing is lost. This is available in λ , π , and ρ , and in ρ it is available twice over, since $@$ makes the representing term quotable in turn. Note that the binder is here reifying the *delivery* locations inside a continuation, which is dual to the reification of the *contact* location forced at the AC pole (Remark 2.12); the two reifications have different targets and should not be run together. Note what this is: the ability of a theory to internalise its own contexts, which is exactly condition (I) of Section 5 seen from the syntax side. The erasure is not a convenience; it is a structural property that some theories have and others do not.

Erasure by collapse. At the AC pole the layer derivative is term-shaped for an entirely different reason: a bag with a hole is just a smaller bag, because the hole carries no information. Nothing has been internalised; something has been forgotten. The two situations look alike on the page and are opposite in character — one is a faithful embedding, the other a quotient.

CCS: the distinction is irreducible. CCS has no binder in the substitutive sense. Restriction hides a name but is never instantiated, and prefix binds nothing, so there is no abstraction to form and no substitution to plug with; erasure by binding is unavailable. And the layer derivatives above show why erasure by collapse does not cover the gap: the prefix layer differentiates to an *action* and the restriction layer to a *name*, neither of which is a term. A CCS context is a path made of actions, names, and residual bags, and that is irreducibly a different type from a CCS process.

This matters for how the present section may be used. The paradigmatic instance of the interaction cut is the one calculus in which the context/term distinction cannot be erased, so no argument in this note may presuppose the erasure. Accordingly the erosion criterion above is stated purely as failure of injectivity of plug, which is meaningful whether or not Ctx_T embeds into T .

Hence reification. When plug ceases to be injective the surface can no longer be read off the context, and must be *reified* as an explicit datum $I : \text{Surf}$ carried by the introduction, with $\text{near}(I, J)$ supplying the match condition the zipper can no longer supply. Reification is forced, and forced exactly at the point injectivity fails. What reification restores is a coordinate system indexed by the reified surface in place of the collapsed one — and with it, labels that once again separate. The nominal surface is, in this reading, precisely the information the derivative could no longer carry, put back by hand so that context labels retain content.

Remark 2.13 (Consequences downstream). Three things in the sequel follow from this subsection rather than being independent stipulations. First, $\text{Ctx}(G)$ of Definition 3.5 is a multicategory of zippers and its members are the transition labels of Section 3.3, so when plug loses injectivity the label set shrinks and the induced bisimulation coarsens; this is Remark 3.8, now derived rather than observed. Second, interfaces carry a surface component (Definition 3.3) for exactly this reason: if the zipper separated positions, sort and stage would suffice. Third, and least expected, erasure by binding is the syntactic face of internal representability: a theory can code its own contexts precisely when it can abstract over a hole, which is why λ , π , and ρ satisfy condition (I) of Definition 5.6 and CCS does not. The separation of Corollary 5.10 and the failure of erasure in CCS are one fact.

2.4 Defect one: preservation is the cheap direction

Proposition 2.14. *For any G and any H possessing an inert term 0 with $0 \not\approx$, the constant map $P \mapsto 0$ is a naïve morphism.*

Proof. Immediate: the hypothesis of the implication is discharged by a constant conclusion. $\square$

The trivial encoding preserves bisimulation because *preservation* is the cheap direction. What one wants is *reflection*, $f(P) \approx_H f(Q) \implies P \approx_G Q$, i.e. faithfulness on the bisimulation quotient. This kills the constant map at once. It does not, however, suffice.

Remark 2.15 (Table lookup). If H is Turing complete, enumerate the terms of G and define f by case analysis on the Gödel index, choosing pairwise non-bisimilar images. The result is faithful and worthless: all the content sits in the metatheoretic map, none in H . Faithfulness must therefore be paired with a condition forbidding the encoding to inspect its argument. The obvious such condition — that f be a homomorphism of the signature — is refuted by the next subsection.

2.5 Defect two: signature preservation does not typecheck

Consider Milner’s encoding of the λ -calculus into π [3]. In the call-by-name presentation,

$$\begin{aligned}\llbracket \lambda x.M \rrbracket p &= p(x).p(q).\llbracket M \rrbracket q, & \llbracket x \rrbracket p &= \bar{x}\langle p \rangle, \\ \llbracket MN \rrbracket p &= (\nu q)(\llbracket M \rrbracket q \mid (\nu y)(\bar{q}\langle y \rangle.\bar{q}\langle p \rangle \mid !y(r).\llbracket N \rrbracket r)).\end{aligned}$$

Two things are visible.

First, $\llbracket - \rrbracket$ is not a function from terms to terms. Its target is name-abstracted processes: $\llbracket - \rrbracket : \Lambda \rightarrow (\mathcal{N} \multimap \Pi)$, each term being encoded relative to a continuation name at which its value is delivered. A signature homomorphism $\Lambda \rightarrow \Pi$ is not merely false here; it is not a well-formed demand.

Second, even granting the parameterisation, no operator of Π corresponds to application. The application clause introduces restriction, replication, and a two-step forwarding protocol. What is uniform about the clause is that a fixed *context* is used, with the encodings of the subterms plugged into its holes — a context-level, not a signature-level, regularity.

Remark 2.16 (The kernel). Nor is the encoding faithful. The equivalence it induces on λ -terms is, by Sangiorgi’s analysis of the lazy encoding [4], Lévy–Longo tree equality; the call-by-value variant induces an eager-normal-form equivalence. Both are strictly coarser than β -conversion. So the canonical example of a good encoding fails both of the obvious repairs: it is not a signature homomorphism, and it is not faithful.

The moral we draw is the one that motivates the whole of Section 3. The point of moving to bisimulation is to break with syntactic constraints; syntactic constraints should therefore re-enter only as optional refinements, never as the primitive. What survives the break is not the signature but the *theory of contexts*.

2.6 Defect three: ambient contexts observe too much

Suppose $\iota : \pi \rightarrow \text{rho}$ is any reasonable encoding. Rho possesses a reflection operator $@$ turning a process into a name, and contexts built from it can inspect the code of their hole in ways no π context can. Consequently there are π -terms $P \approx_\pi Q$ whose images are separated by some rho context. If bisimulation on the target is computed over *all* rho contexts, no encoding of π into rho preserves bisimulation, and the naive category records the fact that rho is more expressive than π as the *absence* of any morphism between them. That is exactly backwards: the extra expressive power of the target should be recorded as data about the encoding, not as an obstruction to its existence.

Remark 2.17 (Two dual failures). Defects two and three are dual and must not be conflated.

- *Under-discrimination*: the encoding identifies source terms the source distinguishes. Milner’s $\lambda \rightarrow \pi$, whose kernel is Lévy–Longo equality.
- *Over-discrimination*: the target’s own contexts separate images the source considers equal. $\pi \rightarrow \text{rho}$.

Restricting the class of observing contexts can only make the target equivalence *coarser*. It therefore repairs over-discrimination completely and under-discrimination never. Two different instruments are required, and Section 4 supplies both: relativisation of the label class for the first, and an image factorisation recording the kernel for the second.

3 Contexts as the primary structure

3.1 Stages

A first attempt at the objects of the context multicategory would take them to be lists of sorts together with the name-arity data recording which free names a term may use. That is the wrong definition, and wrong in the way this note exists to avoid: it presupposes a sort of names with substitution behaviour, which is structure imported from π and ρ rather than structure common to all interactive GSLTs. It is ill-formed for the λ -calculus, where binders bind variables and not channels; it is vacuous for Turing machines; and it is inert for CCS, whose restriction operator hides a name but is never instantiated, so that there is no substitution for the arity data to track. The correct definition is parametric in the binding signature and degenerates, rather than breaks, in each of these cases.

Definition 3.1 (Stage category). The binding signature of G determines a small category $\mathbb{F}_G$ of *stages*: objects are the finite name or variable contexts in which a term of G may be formed, and morphisms are the admissible maps between them. We require that passing beneath a binder be witnessed by an inclusion $n \hookrightarrow n \sqcup \{a\}$, and permit $\mathbb{F}_G$ to be terminal.

Remark 3.2 (Stages across the instances). The stage category, not a clause in the definition of interface, is where the instances differ.

Theory	$\mathbb{F}_G$	Action
TM, finite CCS	terminal	none
CCS with restriction	$\mathbb{I}$: finite sets, injections	renaming; binders hide but never instantiate
λ	free-variable contexts	substitution of terms for variables
π	$\mathbb{F}$: finite sets, all maps	substitution; names may be identified
ρ	fixed point with $\mathcal{N} \cong @Proc$	reflection; the name object is the term object

This is already an expressiveness stratification, and Section 5 will show that some of the separations in the probe order are visible at the level of $\mathbb{F}_G$ alone, without descending to particular terms.

3.2 The context multicategory

Definition 3.3 (Interface). An *interface* of G is a triple (s, n, σ) with $s \in S$ a sort, n an object of $\mathbb{F}_G$, and σ a finite multiset of elements of Srf — the surfaces offered at that position.

Remark 3.4 (Why interfaces carry surfaces). A context can interact with its hole exactly when it offers a surface near, in the sense of Definition 2.5, to one the hole exposes. Recording σ is therefore not bookkeeping: it is what makes hole-filling *interaction-relevant* rather than merely syntactic. Composition of contexts composes surface multisets, with pairs consumed by $\smile$ replaced by their surface of contact. Sort says what kind of thing sits at the position, stage says which names are in scope there, and surface says what interaction is on offer; Section 5 separates theories using the second, and the third is what the labels of Section 3.3 ultimately record.

Definition 3.5 (Context multicategory). The *context multicategory* $\text{Ctx}(G)$ has interfaces as objects. A morphism

$$C : (s_1, n_1, \sigma_1), \dots, (s_k, n_k, \sigma_k) \longrightarrow (s, n, \sigma)$$

is a term of sort s formed at stage n offering surfaces σ , with k labelled holes taken modulo $\equiv$, where hole i has sort s_i , occurs beneath a chain of binders exhibiting a map $n \rightarrow n_i$ in $\mathbb{F}_G$, and is offered the surfaces σ_i by the surrounding term. Composition is hole-filling, defined when the stage and surface data of the filler agree with those recorded at the hole; the identity on (s, n, σ) is the bare hole $[-]$; the symmetric group acts by permuting hole labels.

Remark 3.6 (Capture is data, not a defect). Hole-filling is deliberately *not* capture-avoiding: $(va)[-]$ captures, and it must, since that is what distinguishes it from $[-]$. The map $n \rightarrow n_i$ is exactly the record of which binders the hole sits beneath, and it is what makes composition well defined without α -renaming the hole's names apart. Consequently $\text{Ctx}(G)$ cannot be formed in a setting where substitution is capture-avoiding by fiat; the stage data is what replaces that discipline.

Remark 3.7 (Degeneration). For Turing machines and for finite CCS without restriction, $\mathbb{F}_G$ is terminal, every interface is a sort, and Definition 3.5 reduces to the naive one. For CCS with restriction the stages are non-trivial and they matter: $(va)([-] \mid \bar{a}.0)$ and $[-] \mid \bar{a}.0$ are distinct contexts precisely because their holes sit at different stages. For the λ -calculus, n is the free-variable context. For $C(G)$ the adjoined sorts Sig and Stk enlarge S and leave $\mathbb{F}_G$ untouched, since signatures and the added sorts bind nothing. No instance requires a special case.

Remark 3.8 (Equations on K coarsen the probe). $\text{Ctx}(G)$ is formed modulo $\equiv$. If K carries equations — regime (b) of Remark 2.7 — then two things happen at once, and they are the same thing counted twice. Contexts that were positionally distinct become identified, so $\text{Ctx}(G)$ shrinks and the label class L_i it induces on any target shrinks with it; and $\smile$ is enlarged, so more introductions can meet than position licenses. Loss of positional discrimination in the probe and leakage of ambient authority in the theory are one phenomenon. A theory with a free K is therefore a strictly sharper probe than the same theory with equations imposed on K , and the two presentations sit at different degrees.

Remark 3.9 (Terms are the nullary contexts). A term of sort s at stage n is precisely a morphism $() \rightarrow (s, n)$. Hence a functor $\text{Ctx}(G) \rightarrow \text{Ctx}(H)$ encodes terms and contexts by one and the same map, and their compatibility is functoriality rather than an axiom imposed on top. This is the technical content of the slogan that the contexts must be mapped as well.

Remark 3.10 (Why a multicategory). Parallel composition is genuinely binary and restriction genuinely binds across both arguments, so the interesting contexts have several holes. A category of unary contexts loses exactly the structure that distinguishes concurrent theories. Density in Section 4 is also a multi-ary condition: it asks how ambient contexts decompose, and the decomposition is by parallel assembly.

3.3 Context-labelled transitions

This is the move Section 2.3 was written to prepare. Labels are not drawn from a separately postulated alphabet of actions; they are the contexts themselves, which is to say the locations of Definition 3.5. Everything established there about when locations separate is therefore a statement about how much these labels can distinguish, and the reader should carry it forward: where plug is injective the labelling is faithful, where it is not the labelling collapses, and reification of the surface is what restores it.

Definition 3.11 (Context-labelled transition system). Let $L \subseteq \text{Ctx}_1(G)$ be a class of unary contexts containing $[-]$ and closed under composition. The transition system $\text{CLTS}(G, L)$ has terms as states and

$$P \xrightarrow{C} Q \quad \text{iff} \quad C \in L \text{ and } C[P] \rightarrow Q,$$

with $\Downarrow$ as the atomic predicate on states.

Definition 3.12 (L -bisimulation). $\sim_L$ is the largest symmetric relation R on terms such that $P R Q$ implies (i) $P \Downarrow$ iff $Q \Downarrow$, and (ii) whenever $P \xrightarrow{C} P'$ there is $Q \Rightarrow \xrightarrow{C} Q'$ with $P' R Q'$, where $\Rightarrow$ is reflexive-transitive silent reduction.

Remark 3.13 (Relation to derived congruences). Taking $L = \text{Ctx}_1(G)$ entire makes the label set unmanageably large, which is why Leifer and Milner restrict to contexts minimal for enabling a reaction, via relative pushouts [6, 7, 8]. We note that relativisation to the image of a probe, introduced next, is an *alternative* taming device: if the label class is already the image of a small theory, minimality is not needed to keep it tractable. Whether the two tamings agree on shared examples is Problem 2.

3.4 Morphisms

Definition 3.14 (Ctx-morphism). A *Ctx-morphism* $\iota : G \rightarrow H$ is a symmetric pseudofunctor of multicategories $\iota : \text{Ctx}(G) \rightarrow \text{Ctx}(H)$ such that

- (i) $\iota([-]) \approx [-]$ and $\iota(C \circ (D_1, \dots, D_n)) \approx \iota(C) \circ (\iota D_1, \dots, \iota D_n)$, the comparisons being the structure 2-cells of the pseudofunctor;
- (ii) ι reflects and preserves the primitive observation: $C[\vec{P}] \Downarrow$ iff $\iota(C)[\iota \vec{P}] \Downarrow$;
- (iii) ι is *stage-indexed*: it is accompanied by a functor $\varphi_\iota : \mathbb{F}_G \rightarrow \mathbb{F}_H$ over which it acts on interfaces, $(s, n) \mapsto (\iota s, \varphi_\iota n)$, compatibly with the maps $n \rightarrow n_i$ recorded at holes.

Write **CtxGSLT** for the resulting bicategory, with 2-cells the bisimulations between images.

Note that “pseudo” is not fussiness. Encodings routinely preserve context composition only up to administrative reduction — Milner’s forwarders are exactly that — so strict functoriality would exclude the examples.

Definition 3.15 (Induced label class and relative bisimulation). For $\iota : G \rightarrow H$ put $L_\iota := \iota(\text{Ctx}_1(G))$, closed under composition. Two H -terms X, Y are ι -*indistinguishable*, written $X \sim_\iota Y$, if they are bisimilar in $\text{CLTS}(H, L_\iota)$.

This is the formal content of the observation that when π is encoded into ρ , the encoding may be probed only by the images of π contexts.

Proposition 3.16 (Preservation is now structural). *For any Ctx-morphism ι , the assignment $P \mapsto \iota(P)$ is a morphism of transition systems $\text{CLTS}(G, \text{Ctx}_1(G)) \rightarrow \text{CLTS}(H, L_\iota)$. Consequently $P \sim_{\text{Ctx}_1(G)} Q$ implies $\iota P \sim_\iota \iota Q$.*

Proof. If $P \xrightarrow{C} Q$ then $C[P] \rightarrow Q$; applying ι and using (i) of Definition 3.14, $\iota(C)[\iota P]$ is bisimilar to $\iota(C[P])$, which by (ii) and the fact that pseudofunctors preserve the reduction witnessed by $\Downarrow$ reduces to a term bisimilar to ιQ . Labels are transported because $\iota(C) \in L_\iota$ by definition. The consequence is the standard transport of bisimilarity along a morphism of transition systems. $\square$

Remark 3.17. Proposition 3.16 costs nothing because functoriality was built into Definition 3.14. This is the first dividend of taking contexts as primary: the compositionality condition that had to be imposed as an axiom in the naive setting is now definitional, and the induced kernel (Definition 4.1) is automatically a congruence.

3.5 Adequacy of the probe

Relativising the label class introduces a new way to cheat: the fewer contexts one admits, the coarser the induced equivalence and the easier preservation becomes. In the limit, admitting no contexts makes everything equivalent. The invariant of interest is therefore always the pair (ι, L_ι) , never the encoding alone, and the probe must be pinned down independently.

Definition 3.18 (Self-separating probe). G is *self-separating* if $\sim_{\text{Ctx}_1(G)}$ coincides with the intended weak bisimulation $\approx_G$.

If a probe is not self-separating then a statement of the form “the encoding preserves bisimulation over encoded G contexts” preserves something other than what G means by equivalence, and the whole exercise is void. Self-separation is a hypothesis on the probe alone, verified once per probe, and is the standard question of whether a derived context congruence agrees with barbed congruence [6, 7].

4 The two conditions

4.1 Hosting

Definition 4.1 (Condition (F): hosting). A Ctx-morphism $\iota : G \rightarrow H$ *hosts* G , written $\iota \models (F)$, if it is faithful up to bisimulation: for all parallel C, D in $\text{Ctx}(G)$,

$$\iota(C) \sim_\iota \iota(D) \implies C \sim_{\text{Ctx}_1(G)} D.$$

The *kernel* of ι is the congruence $C \approx_\iota D \iff \iota(C) \sim_\iota \iota(D)$; condition (F) says the kernel is as fine as possible.

Proposition 4.2 (Non-degeneracy). *The constant assignment $\iota(-) = 0$ does not satisfy (F) whenever G has two non-bisimilar terms.*

Proof. All nullary contexts have the same image, so the kernel is the total relation, which is not $\sim_{\text{Ctx}_1(G)}$ under the hypothesis. $\square$

So the degeneracy that motivated the search is excluded by (F) alone, and by a condition that mentions neither syntax nor logic.

4.2 Exhausting

Hosting says the target is at least as expressive as the probe. The complementary question — whether the target is anything *more* — is density.

Definition 4.3 (Restricted nerve). For $\iota : G \rightarrow H$ define

$$N_\iota : \text{Ctx}(H) \longrightarrow \text{Psh}(\text{Ctx}(G)), \quad N_\iota(X)(C) = \text{Sim}(\iota C, X),$$

where $\text{Sim}(-, -)$ is the set of simulations, taken up to $\sim_\iota$.

Definition 4.4 (Condition (D): exhausting). $\iota \models (D)$ if ι is *dense*, i.e. N_ι is fully faithful. Equivalently, every H -context is a canonical colimit of the diagram of ι -images mapping into it.

Proposition 4.5 (Operational reading). $\iota \models (D)$ iff every H -context D is $\sim_\iota$ -indistinguishable from a context assembled out of ι -images: there is a finite diagram $C_1, \dots, C_k$ in $\text{Ctx}(G)$ and an assembly of $\iota C_1, \dots, \iota C_k$ by parallel composition, restriction, and hole-filling that is $\sim_\iota$ -equivalent to D .

Design choice 4.6 (Which colimits). Proposition 4.5 is stated for the class of *assembly colimits*: those generated by parallel composition, restriction, and hole-filling, closed under the reduction-directed joins needed to make the presentation stable under $\rightarrow$. Pinning this class down precisely is the main piece of unfinished business in this note; see Problem 1. We use the operational reading as the working definition throughout and flag every place where the choice matters.

Proposition 4.7 (Composition). *Condition (F) is stable under composition. Condition (D) is stable under composition of morphisms preserving assembly colimits.*

Proof. For (F), faithfulness composes. For (D), if every H -context is an assembly of ι -images and every K -context is an assembly of κ -images, then every K -context is an assembly of assemblies of $\kappa\iota$ -images, and an assembly of assemblies is an assembly provided κ carries the inner assemblies to assemblies — which is the stated hypothesis. $\square$

Remark 4.8. The hypothesis in Proposition 4.7 is not idle: density is famously not stable under composition in general, and the failure is exactly failure of cocontinuity. This is a real constraint on the theory, not a technicality to be assumed away.

4.3 Relative universality

Definition 4.9 (The probe order). Write $G \preceq H$ if some Ctx-morphism $\iota : G \rightarrow H$ satisfies (F); say H is G -universal. Write $G \equiv H$ if some ι satisfies both (F) and (D); say H is G -exact. Write $G \prec H$ if $G \preceq H$ but no ι satisfying (F) also satisfies (D).

Proposition 4.10. *$\preceq$ is a preorder on theories, and $\equiv$ is the induced equivalence on morphisms preserving assembly colimits.*

Proof. Reflexivity by the identity, transitivity by Proposition 4.7. $\square$

So theories carry *degrees* under $\preceq$, and $\prec$ is a strictness relation with a positive criterion: hosted but not exhausted. The observation that arbitrary rho contexts see strictly more than π contexts is now a theorem-shaped assertion — that no (F)-morphism $\pi \rightarrow \text{rho}$ is dense — rather than an obstruction to the existence of morphisms.

4.4 Image factorisation

Condition (F) is too strong to demand of every encoding, since Milner’s fails it. The general statement is a factorisation.

Proposition 4.11 (Image factorisation). *Every Ctx-morphism $\iota : G \rightarrow H$ factors as*

$$G \xrightarrow{q} G / \approx_i \xhookrightarrow{\bar{\iota}} H$$

where q is the quotient by the kernel and $\bar{\iota} \models (F)$.

Proof. The kernel $\approx_i$ is a congruence by Remark 3.17, so the quotient multicategory exists; ι factors through it by construction, and the induced $\bar{\iota}$ is faithful by definition of the kernel. $\square$

Example 4.12 (Milner, located). For Milner’s call-by-name encoding, q is the quotient of Λ by Lévy–Longo tree equality and $\bar{\iota}$ hosts the quotient inside π . The encoding is thus completely described by the pair: a nontrivial quotient recording under-discrimination, followed by a hosting embedding. Neither half is visible in Definition 2.11.

Remark 4.13 (The full invariant). An encoding is characterised by three data: the kernel (how much of the source is lost), the failure of density (how much of the target is unseen), and — once Section 6.4 is in play — the overhead. Reporting only that “an encoding exists” discards all three.

5 Refining Turing completeness

5.1 The classical notion as a trace collapse

Definition 5.1 (Trace collapse). The *trace collapse* of $\mathbf{CtxGSLT}$ is the bicategory obtained by replacing the 2-cells — bisimulations — by trace inclusions, so that two terms are identified when they admit the same sequences of labels.

By Remark 2.3 the Turing machine is not an object of the category, so before it can serve as a probe at all it must be *presented* as one.

Definition 5.2 (A presentation of TM). A *process presentation* of Turing machines is an interactive GSLT $\mathbf{TM}^\sharp$ whose interacting sort contains both automaton and tape — for instance by encoding the tape as a process — with K the constructor placing an automaton beside a tape, and whose interaction rule is the machine’s transition relation.

Such a presentation exists, and that is the whole difficulty: it is a choice. Different encodings of the tape yield different $\mathbf{Ctx}(\mathbf{TM}^\sharp)$ and hence different probes, and nothing internal to the machine model selects among them. The classical yardstick is therefore not merely coarse; it is not even well defined as a probe until an extraneous decision has been made.

Proposition 5.3 (Recovering Turing completeness). *Fix a process presentation $\mathbf{TM}^\sharp$. In the trace collapse, H is $\mathbf{TM}^\sharp$ -universal if and only if H is Turing complete in the classical sense, and the criterion is independent of which presentation was fixed.*

Proof sketch. In the trace collapse the branching structure of $\mathbf{Ctx}(\mathbf{TM}^\sharp)$ is invisible, so the only surviving content of faithfulness on the nullary component is that distinct input/output functions have distinct images — i.e. that H realises every partial recursive function. Conversely a universal machine simulated in H supplies the morphism. Independence of the presentation holds because any two process presentations of the same machine model compute the same functions, and at trace level that is all that is being compared. $\square$

Remark 5.4 (Presentation-dependence is the diagnostic). The independence clause of Proposition 5.3 holds *only* in the trace collapse. Before collapsing, distinct process presentations of the same machine model sit at distinct degrees, since they induce distinct context multicategories. The classical notion is thus well defined precisely to the extent that it is blind, and the coincidence of those two facts is not an accident: what makes it insensitive to the choice of presentation is exactly what makes it insensitive to branching.

Remark 5.5 (Why the classical notion is blind). Above the Turing threshold the trace-collapsed order has a single degree: any two Turing-complete theories host one another at trace level. The branching-time order does not collapse, and the separation results [9, 10, 11] are exactly exhibitions of that non-collapse. Classical Turing completeness is thus the shadow of $\preceq$ under Definition 5.1, and using it as a hypothesis for a bisimulation-level conclusion is using the shadow to constrain the object.

5.2 The ladder

The following stratification replaces the single predicate. Each level adds one requirement; the classical notion is level 0, and level 4 is what a genuine claim of behavioural universality amounts to.

Level	Condition	Comment
0	functional universality	classical Turing completeness; trace level
1	sound simulation	ι is a Ctx-morphism; no faithfulness claimed
2	hosting (F)	kernel is trivial; source runs faithfully in target
3	internal representability (I)	the encoding is realised by a single target term applied to codes, not by a metatheoretic family
4	exhausting (D)	target adds nothing the source cannot assemble

Definition 5.6 (Condition (I): internal representability). $\iota \models (I)$ if there is a coding $\ulcorner - \urcorner$ of G -terms as H -data, definable in H , and a single H -term U_i with $U_i \langle \ulcorner P \urcorner \rangle \sim_i \iota(P)$ for every G -term P .

Condition (I) is what turns a translation into an interpreter, and it factors exactly along the ladder of Section 2.

Proposition 5.7 (Continuity and reflection give (I)). *Let G and H be continued (Definition 2.6) and let H be reflective, in the sense that it carries a term-level quoting operator $@$ with $*@P \equiv P$. Then any Ctx-morphism $\iota : G \rightarrow H$ satisfies (I).*

Proof sketch. By clause (ii) and Proposition 2.9, $\text{Ctx}(G)$ admits codes, so $\ulcorner - \urcorner$ is a computable map from G -terms to finite data. By reflection of H , that data is representable as an H -name, and $*$ recovers the process it names; U_i is then the H -term that decodes and dispatches, definable because the guards it must decide are decidable by clause (ii) of H . $\square$

The two hypotheses do distinct work, and separating them is the point: continuity of G supplies canonical representatives to be coded, reflection of H supplies an internal home for the codes. In π the second fails — processes are not transmissible data, so any coding is metatheoretic — while in ρ both hold by construction. Non-naturality of an internalisation family and externality of its coding are the same defect seen twice, and Proposition 5.7 says exactly which hypothesis repairs it.

5.3 Separations

Probe / target	(F)	(I)	(D)	Obstruction
$\text{TM}^\sharp \rightarrow \text{CCS+rec}$	✓	✓	—	no name passing; probe needs a presentation
$\pi \rightarrow \text{CCS+rec}$	—	—	—	fixed interface
$\lambda \rightarrow \pi$	—	✓	—	kernel is Lévy–Longo
$\pi \rightarrow \rho$	✓	✓	—	@ not assemblable
$\rho \rightarrow \rho$	✓	✓	✓	—

5.4 Separation at the level of stages

The entry for $\pi \rightarrow \text{CCS}$ deserves its own statement, since it is the cleanest separation the order provides and it is invisible classically. With Definition 3.1 in hand it can be argued at the level of stage categories, without descending to a particular pair of terms.

Definition 5.8 (Stage-changing transition). A transition $P \xrightarrow{C} Q$ is *stage-changing* if the stage of Q properly extends that of P in $\mathbb{F}_G$.

Bound output in π is stage-changing: extruding a private name enlarges the observer's stage. Restriction in CCS is not: $(\nu a)P$ never emits a , so every CCS transition is stage-preserving, and $\mathbb{F}_{\text{CCS}} = \mathbb{I}$ contains only injections.

Proposition 5.9 (Hosting constrains the stage functor). *If $\iota : G \rightarrow H$ satisfies (F), then $\varphi_\iota : \mathbb{F}_G \rightarrow \mathbb{F}_H$ is faithful and preserves the substitution action: for every map σ in $\mathbb{F}_G$ whose induced action separates two G -terms, $\varphi_\iota \sigma$ separates their images.*

Proof sketch. Suppose $\varphi_\iota \sigma = \varphi_\iota \tau$ for distinct σ, τ separating P from Q . The G -contexts realising σ and τ then have $\sim_\iota$ -equal images, so the kernel of ι identifies them, contradicting (F). Preservation of the action is the same argument applied to the nullary component. $\square$

Corollary 5.10 ($\pi \not\leq \text{CCS}$). *No Ctx-morphism $\pi \rightarrow \text{CCS} + \text{rec}$ satisfies (F), although $\text{CCS} + \text{rec}$ is Turing complete.*

Proof sketch. $\mathbb{F}_\pi$ contains non-injective maps: a π context may deliver a name that identifies two names previously distinct, turning an inert composition into a redex. Every map of $\mathbb{F}_{\text{CCS}} = \mathbb{I}$ is injective, so φ_ι must send a non-monic to a monic and therefore identifies maps that $\mathbb{F}_\pi$ distinguishes. By Proposition 5.9 this contradicts (F). $\square$

Remark 5.11 (Residual hypothesis). The sketch rules out encodings that are stage-indexed in the sense of Definition 3.14(iii). It does not by itself rule out an encoding that pre-allocates a fixed finite pool of CCS channels and simulates identification by case analysis over the pool. Excluding those requires the additional hypothesis that φ_ι be finitary and uniform — essentially the finite-support condition of nominal sets. We record this as a hypothesis rather than pretending it is discharged; it is the same gap that the uniformity criteria of [10] are designed to close, and it is the one place where this note's machinery does not yet subsume the checklist.

Behavioural universality therefore forces name passing; under (I) it forces the coding to land in the name sort, which is reflection.

6 Two meta-circular interpreters

We come to the structural payoff. Conditions (F) and (D) look like two unrelated technical hygiene requirements. They are in fact the two halves of mutual meta-circularity: one gives an interpreter running the source inside the target, the other an interpreter running the target inside the source.

6.1 The source, inside the target

Theorem 6.1 (Hosting yields an interpreter for the source). *Let $\iota : G \rightarrow H$ satisfy (F) and (I), and let G be self-separating. Then U_ι is an interpreter for G written in H , correct up to $\sim_\iota$: for every G -term P and every G -context C ,*

$$U_\iota \langle \ulcorner C[P] \urcorner \rangle \sim_\iota \iota(C) [U_\iota \langle \ulcorner P \urcorner \rangle],$$

and $P \approx_G Q$ iff $U_\iota \langle \ulcorner P \urcorner \rangle \sim_\iota U_\iota \langle \ulcorner Q \urcorner \rangle$.

Proof. The displayed equation is Definition 5.6 composed with functoriality (i) of Definition 3.14. The right-to-left direction of the second claim is Proposition 3.16; the left-to-right direction is (F) together with self-separation of G , which identifies $\sim_{\text{Ctx}_1(G)}$ with $\approx_G$. $\square$

Write $E_{G \rightarrow H} := U_\iota$ for this interpreter.

6.2 The target, inside the source

Theorem 6.2 (Exhausting yields an interpreter for the target). *Let $\iota : G \rightarrow H$ satisfy (D) with G and H continued. Then the assembly presentations of Proposition 4.5 are effective, and there is a G -term V_ι interpreting H inside G : for every H -context D and H -term X ,*

$$V_\iota \langle \ulcorner D \urcorner, \ulcorner X \urcorner \rangle \approx_G \text{ the } G\text{-context assembly } \delta(D) \text{ applied to } \delta(X),$$

where δ is the computable procedure taking a code for an H -context to a finite assembly of ι -images $\sim_\iota$ -equivalent to it.

Proof sketch. By (D), each H -context is $\sim_\iota$ -equivalent to an assembly of ι -images. By clause (ii) for H and Proposition 2.9, H -contexts admit codes, and by clause (ii) for G the search for an assembly ranges over an effectively enumerable set of G -context representatives with decidable equality, so δ is computable; by faithfulness of ι on the assembled pieces — which is the fully faithful half of Definition 4.4 — the G -contexts underlying those images are determined up to $\sim_{\text{Ctx}_1(G)}$. V_ι therefore decodes, recovers the underlying G -contexts, runs them under G 's own dynamics, and reassembles. Correctness up to $\approx_G$ follows from self-separation of G and Proposition 3.16 applied in the reverse direction along the fully faithful nerve. $\square$

Write $E_{H \rightarrow G} := V_\iota$.

Remark 6.3 (Where the asymmetry lives). Theorem 6.1 needs (I) and Theorem 6.2 needs effectiveness of δ . Both are “the encoding is not merely a metatheoretic function” conditions, imposed once on each side, and both are now consequences rather than hypotheses: (I) follows from continuity together with reflection of the target (Proposition 5.7), and effectiveness of δ follows from the sections of both theories (Proposition 2.9). This is the whole purpose of rung three of Section 2. In a reflective continued theory the same operator supplies both.

6.3 The two round trips

Corollary 6.4 (Two meta-circular interpreters). *Suppose G and H are continued and self-separating, H is reflective, and $\iota : G \rightarrow H$ satisfies (F) and (D). Then (I) and effective assembly hold by Propositions 5.7 and 2.9, and*

$$M_G := E_{H \rightarrow G} \circ E_{G \rightarrow H} : G \rightsquigarrow G, \quad M_H := E_{G \rightarrow H} \circ E_{H \rightarrow G} : H \rightsquigarrow H$$

are interpreters for G in G and for H in H respectively, satisfying

$$M_G \approx_G \text{id}_G, \quad M_H \sim_\iota \text{id}_H.$$

M_G is the source looping through the target; M_H is the target looping through the source.

Proof. Composition of the two theorems. M_G takes a G -term to its code, runs it under H 's dynamics as simulated by V_ι , and returns; each leg preserves and reflects the relevant equivalence, so the composite is behaviourally the identity. Symmetrically for M_H , with $\sim_\iota$ in place of $\approx_H$ because H is observed only through L_ι . $\square$

Definition 6.5 (Meta-circularity). An endo-interpreter $M : G \rightsquigarrow G$ is *meta-circular* if it is syntactically non-trivial — $M(P) \not\equiv P$ for some P — and behaviourally the identity, $M \approx \text{id}$.

Corollary 6.4 says that exactness of a probe relation manufactures a meta-circular interpreter on each side. This is the precise sense in which the two conditions belong together: (F) alone gives a one-way hosting with no return journey, (D) alone gives a decomposition with nothing to decompose faithfully, and only the conjunction closes the loop.

Remark 6.6 (Monad and comonad). $E_{G \rightarrow H} \dashv E_{H \rightarrow G}$ is an adjunction in the simulation bicategory whose induced monad on G is M_G and whose induced comonad on H is M_H . Both are idempotent up to bisimulation, so the adjunction is an adjoint equivalence. In such situations the degeneracy is usually noted with unease — if both unit and counit are invertible, the word “adjunction” does no work. The reading offered here is the opposite: the degeneracy is the content. An adjoint equivalence between two theories in the simulation bicategory is precisely the statement that each can meta-circularly interpret the other. What is missing in such cases is not non-degeneracy but a grading, supplied next.

6.4 Grading: meta-circularity is not free

Definition 6.7 (Graded interpreter). E has *grade* $g : \mathbb{N} \rightarrow \mathbb{N}$ if every labelled step of the source is matched by at most $g(n)$ steps of the target, n the size of the argument. Write $G \preceq_g H$ for hosting with grade g .

Proposition 6.8 (The cost of the loop). *If $E_{G \rightarrow H}$ has grade g and $E_{H \rightarrow G}$ has grade h , then M_G has grade $g \circ h$. Hence $M_G \approx \text{id}_G$ but $M_G \not\approx_1 \text{id}_G$ whenever $g \circ h$ is not the identity: the round trip is behaviourally invisible and resource-visible.*

Corollary 6.9 (Graded lax coreflection). *In the graded simulation bicategory, $E_{G \rightarrow H} \dashv E_{H \rightarrow G}$ is a lax coreflection: the unit has grade 1 and the counit has grade $g \circ h$. The failure to be an equivalence is exactly the interpretive overhead.*

Ungraded, the adjunction collapses and says nothing; graded, its counit is a measurement, and the thing being measured is the price of self-interpretation. The grading is stated here because it costs nothing to state and because the interpretive overhead it names is the only quantity in this note with a plausible claim to being measured rather than merely compared; nothing else below depends on it.

7 Open problems

Problem 1 (Assembly colimits). Pin down the class of colimits in Design 4.6. Density is only as sharp as the colimits one admits, and admitting too many trivialises (D) while admitting too few makes it unsatisfiable. The candidate is the class generated by parallel composition, restriction, and hole-filling, closed under reduction-directed joins; what must be checked is that this class is stable under Ctx-morphisms preserving $\Downarrow$.

Problem 2 (Self-separation of π). Is π self-separating in the sense of Definition 3.18? Equivalently, does the context-labelled bisimulation over all unary π contexts agree with barbed congruence? If the RPO-derived labels of [6] suffice, does the relativised label class L_l agree with them on shared examples?

Problem 3 (The rho probe of λ). Compute the kernel of the canonical encoding $\lambda \rightarrow \text{rho}$. Since reflection permits the tree structure of a term to be internalised, the kernel may be strictly finer than Lévy–Longo. If it is β -conversion itself, that is a sharp statement of what reflection adds over name passing.

Problem 4 ($\pi \prec \text{rho}$). Exhibit a rho context — $@$, or the nearness operator of Definition 2.5 — and prove that it is not $\sim_l$ -equivalent to any assembly of encoded π contexts. This is the first genuine strictness result the order should deliver, and it is the one the classical yardstick is constitutionally unable to see.

Remark 3.2 suggests attacking it at the level of stages rather than terms, as Corollary 5.10 does for the CCS case: $\mathbb{F}_\pi$ is a category of finite sets, whereas $\mathbb{F}_{\text{rho}}$ satisfies $\mathcal{N} \cong @ \text{Proc}$ and

is therefore a fixed point no functor from finite sets can reach. If the argument goes through, both separations become statements about binding signatures alone, and the probe order acquires a computable lower bound.

Problem 5 (Discharging Remark 5.11). Formulate the finitary uniformity hypothesis on φ_i precisely — most likely as finite support in the sense of nominal sets — and verify that it holds for every Ctx-morphism arising from an internally representable encoding in the sense of Definition 5.6. If (I) implies it, the residual gap closes and the separation arguments become unconditional.

Problem 6 (Well-foundedness of the graded order). Is $\preceq_g$ well-founded? A descending chain of theories each hosting the next with strictly better grade would be a hierarchy of increasingly efficient self-interpretation, and its existence or absence is the quantitative analogue of the degree structure.

Problem 7 (Probe profiles). A theory's invariant under this analysis is not a single degree but the profile of its kernels and densities across a family of probes. Is the profile of ρ determined by finitely many probes?

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